

PHRAE, Thailand

WMO: 483300

Lat: 18.1286N

Lon: 100.1625E

Elev: 163

StdP: 99.38

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.2	14.7	10.2	7.9	20.8	11.8	8.8	21.1	4.3	29.5	3.8	29.3	0.1	10	0.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	12.1	38.8	25.2	37.4	25.0	36.2	25.0	27.9	33.4	27.4	32.9	27.1	32.4	2.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.5	30.4	26.1	22.0	29.9	25.8	21.5	29.4	90.4	33.6	88.3	32.9	86.6	32.6	33.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.8	4.2	3.5	DB	10.8	40.2	2.0	1.4	9.4	41.2	8.2	42.0	7.1	42.8	5.7	43.8	(4)
(5)			WB	10.2	28.8	1.9	1.3	8.9	29.8	7.8	30.5	6.7	31.2	5.3	32.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	23.6	25.7	28.6	30.8	29.9	29.1	28.4	28.1	28.2	27.8	26.0	23.7	(6)	
	DBStd	2.83	2.03	1.92	2.07	2.01	2.14	1.40	1.35	1.16	1.07	1.35	1.99	2.42	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	2	1	0	0	0	0	0	0	0	0	0	0	1	(9)	
	CDD10.0	6382	423	440	576	624	616	574	569	560	547	550	479	424	(10)	
	CDD18.3	3342	165	207	318	374	357	324	311	302	297	292	229	167	(11)	
	CDH23.3	35182	1401	2063	3749	4847	4292	3704	3280	3002	2907	2687	1911	1339	(12)	
(13)	CDH26.7	15998	576	1042	2102	2813	2173	1608	1234	1032	1042	1059	797	519	(13)	
(14) Wind	WSAvg	0.8	0.5	0.7	1.0	1.2	1.1	1.2	1.1	0.9	0.6	0.5	0.4	0.5	(14)	
(15) Precipitation	PrecAvg	1140	9	4	25	53	174	122	144	293	191	103	20	4	(15)	
	PrecMax	1558	71	17	107	101	353	285	293	452	283	224	76	19	(16)	
	PrecMin	804	0	0	0	0	51	45	74	182	70	24	4	0	(17)	
	PrecStd	218	20	7	30	38	83	58	57	94	65	57	22	6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	36.2	38.9	41.1	40.5	37.0	35.8	34.1	34.8	35.2	34.8	34.0	(19)	
		MCWB	22.8	23.0	23.6	25.1	25.1	26.0	26.5	27.1	27.2	26.7	24.9	23.9	(20)	
	2%	DB	32.5	35.1	37.6	39.2	37.9	35.2	34.0	33.1	33.4	33.6	33.4	32.2	(21)	
		MCWB	22.2	22.8	23.7	25.4	26.0	26.4	26.5	26.7	26.8	26.3	24.6	23.2	(22)	
	5%	DB	31.4	34.0	36.5	37.9	36.2	34.0	32.9	32.2	32.5	32.8	32.2	31.1	(23)	
		MCWB	21.8	22.7	23.7	25.4	26.1	26.3	26.3	26.3	26.5	26.0	24.2	22.8	(24)	
	10%	DB	30.2	32.8	35.2	36.5	34.7	32.9	31.8	31.2	31.5	31.8	31.1	29.8	(25)	
		MCWB	21.4	22.5	23.7	25.4	26.1	26.1	26.0	26.1	26.3	25.7	23.8	22.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.0	25.6	26.9	28.0	28.4	28.0	27.9	28.1	28.2	28.0	26.6	25.6	(27)	
		MCDB	31.1	33.5	35.7	36.4	34.3	33.9	32.8	32.5	32.7	33.0	31.8	31.3	(28)	
	2%	WB	23.8	24.7	25.9	27.1	27.6	27.2	27.1	27.3	27.5	27.1	25.8	24.4	(29)	
		MCDB	29.4	32.1	34.3	35.7	33.8	32.6	32.0	31.6	31.8	32.1	31.1	29.9	(30)	
	5%	WB	23.0	24.0	25.2	26.5	27.1	26.8	26.7	26.8	27.0	26.6	25.2	23.6	(31)	
		MCDB	28.7	31.3	33.4	35.0	33.3	31.9	31.2	30.8	31.3	31.2	30.1	28.8	(32)	
	10%	WB	22.2	23.3	24.6	26.0	26.6	26.5	26.2	26.4	26.5	26.1	24.7	22.9	(33)	
		MCDB	27.9	30.3	32.4	34.2	32.5	31.2	30.4	30.1	30.4	30.4	29.0	27.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	15.1	14.0	12.1	9.3	7.8	6.9	6.7	7.4	8.7	10.8	12.9	(35)	
		MCDBR	15.6	16.4	15.6	13.4	11.7	9.4	8.7	8.1	8.7	9.9	12.0	13.5	(36)	
		MCWBR	6.1	5.9	4.9	3.9	3.3	2.9	3.0	3.1	3.3	3.7	4.3	5.2	(37)	
	5% WB	MCDBR	13.0	14.2	13.8	12.1	9.8	8.3	7.8	8.1	9.0	10.3	12.0	12.0	(38)	
		MCWBR	5.3	5.4	4.9	4.1	3.5	3.0	3.0	3.1	3.4	3.8	3.9	4.8	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.455	0.553	0.674	0.645	0.500	0.446	0.437	0.451	0.486	0.530	0.444	0.444	(40)	
	taud		2.156	1.908	1.656	1.771	2.191	2.368	2.388	2.330	2.218	2.051	2.264	2.213	(41)	
	Ebn at Noon		821	758	683	708	809	846	855	849	821	768	828	819	(42)	
	Edh at Noon		143	193	256	230	149	124	121	130	144	165	128	132	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.57	5.18	5.43	5.93	5.43	4.87	4.51	4.43	4.62	4.65	4.72	4.39	(44)	
	RadStd		0.24	0.21	0.32	0.34	0.47	0.37	0.33	0.28	0.23	0.25	0.26	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+180	+182	(46)	
(47)	Regional (20 neighbors)	+0.26	N/A	N/A	N/A	N/A	+0.12	N/A	N/A	+125	+126	(47)	

Nomenclature: See separate page